

HEALTHCONNECT ADVANCED ANALYTICS & DECISION SUPPORT REPORT

Week 6 - Data Analytics Track

AnalystLab Africa - Experience Lab

1. EXECUTIVE SUMMARY

This report presents advanced analytics for HealthConnect Clinic, building on Week 5 analysis. Key findings have been validated using statistical testing and integrated with the Data Science track.

2. ADVANCED ANALYSIS

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2.1 SEGMENTATION ANALYSIS

No-Show Rate by Appointment Type:

Appointment Type	No-Show Rate (%)	Count
General Consultation	[value]	[count]
Specialist Consultation	[value]	[count]
Follow-up	[value]	[count]
Diagnostic Test	[value]	[count]

Key Finding: [Highest type] has the highest no-show rate

No-Show Rate by Age Group:

Age Group	No-Show Rate (%)	Count
18-24	[value]	[count]
25-34	[value]	[count]

35-44	[value]	[count]
45-54	[value]	[count]
55-64	[value]	[count]
65+	[value]	[count]

Key Finding: [Age group] has the highest no-show rate

No-Show Rate by Gender:

Gender	No-Show Rate (%)	Count
Female	[value]	[count]
Male	[value]	[count]
Prefer not to say	[value]	[count]

Key Finding: [Gender] shows slightly higher no-show rate

2.2 RELATIONSHIP ANALYSIS

Previous No-Shows vs Current No-Show:

Previous No-Shows	No-Show Rate (%)	Count
0	[value]	[count]
1	[value]	[count]
2	[value]	[count]
3+	[value]	[count]

Key Finding: Strong correlation between previous and current no-shows

Booking Lead Days vs No-Show:

Booking Lead Days	No-Show Rate (%)	Count
0-7	[value]	[count]
7-14	[value]	[count]
14-30	[value]	[count]
30-60	[value]	[count]
60+	[value]	[count]

Key Finding: Longer booking lead times = higher no-show rates

3. STATISTICAL VALIDATION

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3.1 CHI-SQUARE TEST: APPOINTMENT TYPE VS NO-SHOW

Chi-Square Statistic: [value]
P-value: [value]
Significant ($p < 0.05$): [Yes/No]

Conclusion: [Appointment type has/has no significant relationship with no-shows]

3.2 T-TEST: DISTANCE VS NO-SHOW

T-Statistic: [value]
P-value: [value]
Significant ($p < 0.05$): [Yes/No]

Conclusion: [Distance has/has no significant relationship with no-shows]

3.3 CORRELATION ANALYSIS

Variable	Correlation with No-Show
previous_no_shows	[value]
booking_lead_days	[value]
waiting_time_minutes	[value]
distance_to_clinic_km	[value]
age	[value]

Key Finding: [Variable] has strongest correlation

4. KPI VALIDATION

KPI	Value	Status
No-Show Rate (original)	48.0%	Validated
No-Show Rate (excl. Cancelled)	[value]%	More accurate
Reminder Effectiveness	[value]%	Validated
High-Risk Patient Rate	[value]%	Validated

5. EVIDENCE-BASED INSIGHTS

INSIGHT 1: Previous No-Shows are Strongest Predictor

Evidence: Patients with 2+ previous no-shows have [value]% no-show rate

Implication: Target these patients with proactive outreach

INSIGHT 2: Booking Lead Time Matters

Evidence: Longer booking lead times correlate with higher no-shows

Implication: Offer shorter wait times or same-week appointments

INSIGHT 3: Reminders Work

Evidence: Reminders reduce no-shows by [value]%

Implication: Expand reminder programme to all patients

INSIGHT 4: Certain Days Have Higher No-Shows

Evidence: [Day] has highest no-show rate at [value]%

Implication: Optimize scheduling on high-risk days

INSIGHT 5: Distance is a Barrier

Evidence: No-show patients travel [value] km vs [value] km for attended

Implication: Consider transport support for distant patients

6. BUSINESS RECOMMENDATIONS

RECOMMENDATION 1: Target High-Risk Patients

Priority: High

Evidence: Patients with 2+ previous no-shows have [value]% no-show rate

Action: Implement proactive outreach for patients with 2+ previous no-shows

Expected Impact: Reduce overall no-show rate by 10-15%

RECOMMENDATION 2: Reduce Booking Lead Time

Priority: High

Evidence: Longer booking lead times correlate with higher no-shows

Action: Offer shorter wait times or same-week appointments

Expected Impact: Reduce no-show rate by 5-10%

RECOMMENDATION 3: Expand Reminder Programme

Priority: Medium

Evidence: Reminders reduce no-shows by [value]%

Action: Send reminders via multiple channels

Expected Impact: Reduce no-show rate by additional 5%

RECOMMENDATION 4: Optimize Appointment Scheduling

Priority: Medium

Evidence: Certain days have higher no-show rates

Action: Schedule more appointments on low no-show days

Expected Impact: Better slot utilization

RECOMMENDATION 5: Transport Support for Far Patients

Priority: Low

Evidence: No-show patients travel further

Action: Partner with transport services for patients > 15km

Expected Impact: Reduce no-shows for distant patients

7. UPDATED VISUALIZATIONS

1. No-Show by Appointment Type (Bar Chart)
2. No-Show by Age Group (Bar Chart)
3. Previous No-Shows vs No-Show Rate (Bar Chart)
4. Booking Lead Days vs No-Show (Bar Chart)

8. DATA LIMITATIONS

1. Limited patient satisfaction data
2. Missing demographic information
3. No external data (weather, traffic)
4. Some distance values missing

9. CROSS-TRACK CONTRIBUTION

Provided to Data Science:

- Validated KPIs and segment analysis
- Feature recommendations
- Statistical validation results

Received from Data Science:

- Feature importance from model
- Model performance insights
- Confirmation of key predictors

10. WEEK 7 ANALYTICAL TESTING REQUIREMENTS

1. Validate all KPIs with fresh data
2. Test dashboard interactivity
3. Verify statistical significance
4. Test segmentation logic

5. Prepare for Week 8 presentation

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